

VERKHOTURYE, Russian Federation

WMO: 281440

Lat: 58.867N

Lon: 60.783E

Elev: 126

StdP: 99.83

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.0	(c) -31.1	(d) -36.9	(e) 0.1	(f) -33.8	(g) -33.9	(h) 0.2	(i) -31.0	(j) 6.4	(k) -10.5	(l) 5.7	(m) -10.7	(n) 1.1	(o) 20	(p) 0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.2	20.1	27.4	19.0	25.5	17.9	21.1	27.4	20.1	26.0	19.0	23.9	2.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	23.9	17.8	13.0	22.9	16.9	12.2	21.7	61.6	27.7	58.0	25.9	54.5	24.0	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.9	5.8	5.1	-37.8	31.8	2.7	1.5	-39.7	32.9	-41.3	33.7	-42.8	34.6	-44.7	35.6
			-37.6	22.3	2.8	1.2	-39.6	23.2	-41.3	23.9	-42.8	24.5	-44.9	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-15.7	-13.7	-5.0	2.9	10.1	15.4	17.4	14.7	8.9	2.4	-7.5	-13.1
	DBStd	13.08	7.72	7.95	5.66	5.24	5.14	4.52	3.96	4.12	4.44	4.98	8.11	8.43
	HDD10.0	3777	798	663	464	218	65	8	2	6	72	242	524	716
	HDD18.3	6226	1056	897	722	463	259	110	65	128	283	495	774	974
	CDD10.0	670	0	0	0	6	67	170	233	151	39	4	0	0
	CDD18.3	77	0	0	0	0	3	21	37	15	1	0	0	0
	CDH23.3	926	0	0	0	1	77	255	423	161	9	0	0	0
	CDH26.7	228	0	0	0	0	8	58	110	51	1	0	0	0
	Wind	WSAvg	2.2	1.9	2.1	2.5	2.5	2.5	2.3	1.9	2.1	2.4	2.2	2.0
(15) Precipitation	PrecAvg	581	29	19	26	29	52	77	95	86	66	39	35	28
	PrecMax	718	62	45	52	71	102	156	188	158	142	75	61	52
	PrecMin	470	5	4	1	2	14	18	9	22	11	9	5	11
	PrecStd	69	13	11	15	17	25	38	45	39	32	19	16	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.3	9.6	21.7	27.8	30.2	31.6	31.0	25.1	18.1	7.7	3.0
		MCWB	0.9	0.9	4.0	12.0	16.1	19.7	21.1	20.7	15.9	11.9	5.7	1.6
	2%	DB	-0.5	1.3	6.5	17.0	25.8	28.4	29.4	27.6	21.6	14.8	5.6	1.6
		MCWB	-1.6	-0.3	2.3	9.6	15.4	19.1	20.6	19.8	14.7	10.0	4.0	0.1
	5%	DB	-2.7	-0.4	4.5	14.1	23.1	26.5	27.7	24.4	18.3	12.0	3.3	0.0
		MCWB	-3.6	-1.9	1.0	7.9	13.8	18.2	20.1	18.2	13.4	8.6	2.0	-1.2
	10%	DB	-4.9	-2.3	2.7	11.6	20.3	24.3	25.7	22.2	15.8	9.5	1.6	-1.9
		MCWB	-5.8	-3.8	0.1	6.2	12.6	17.0	18.8	17.0	12.3	7.1	0.4	-3.1
	0.4%	WB	0.8	1.4	5.1	12.6	17.7	21.6	22.6	21.6	17.2	12.2	6.1	1.7
		MCDB	1.8	2.8	8.3	20.1	24.9	27.7	28.5	28.9	22.7	16.2	7.3	2.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.5	-0.2	2.9	10.2	16.1	20.2	21.5	20.1	15.5	10.6	4.2	0.3
		MCDB	-0.5	1.5	5.8	16.4	23.8	26.5	27.7	26.4	19.7	14.1	5.6	1.5
	5%	WB	-3.5	-1.9	1.4	8.2	14.8	18.9	20.6	18.9	14.0	8.8	2.1	-1.1
		MCDB	-2.7	-0.4	3.8	13.1	21.9	24.9	26.5	23.6	17.5	12.0	3.1	0.0
	10%	WB	-5.8	-3.8	0.3	6.6	13.2	17.7	19.6	17.7	12.7	7.2	0.5	-2.9
		MCDB	-5.0	-2.4	2.5	11.1	19.3	23.3	24.8	21.4	15.5	9.5	1.5	-2.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	9.9	10.2	10.2	12.2	10.8	11.0	9.7	8.0	6.6	6.2	7.0
		MCDBR	7.0	8.1	11.1	15.0	16.9	14.5	14.4	14.0	11.7	10.9	5.7	5.3
		MCWBR	6.6	6.7	7.6	9.0	8.3	6.5	6.6	6.8	6.2	6.6	4.7	5.0
	5% WB	MCDBR	6.9	7.7	9.1	13.8	15.2	13.2	13.3	12.9	10.2	10.2	5.5	5.6
		MCWBR	6.6	6.5	6.6	8.6	8.3	6.6	6.5	6.5	6.0	6.5	4.7	5.4
(40) Clear-Sky Solar Irradiance	taub		0.275	0.306	0.362	0.401	0.400	0.398	0.416	0.427	0.376	0.338	0.293	0.266
	taud		2.030	2.047	1.974	2.089	2.273	2.337	2.290	2.227	2.329	2.301	2.127	1.980
	Ebn at Noon		503	683	745	793	829	837	809	764	747	658	484	368
	Edh at Noon		53	86	126	134	120	115	118	117	90	68	48	39
(44) All-Sky Solar Radiation	RadAvg		0.50	1.39	2.66	3.96	4.97	5.33	5.33	3.94	2.37	1.18	0.52	0.28
	RadStd		0.06	0.16	0.24	0.46	0.43	0.42	0.48	0.40	0.26	0.18	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page